



Specialized Topics of Theoretical Physics

Majorana Modes in the Machine – Simulating Topological Phases with Quantum Circuits and AI

NISQ Noise on the Diagnostic

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1 Introduction

The 1D Kitaev chain hosts unpaired Majorana zero modes at its ends in the topological phase [1]. In the preceding part of the project, a parity-constrained variational quantum eigensolver (VQE) method prepared the finite-chain ground state and measured the non-local edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$. However, that ideal result does not yet answer the practical question of whether the diagnostic remains visible after imperfect gates, decoherence, readout errors, and a noisy variational loop.

This report concerns itself with *NISQ Noise on the Diagnostic*. The first test we performed kept the validated state frozen and focused on determining readout error, local depolarization, and finite measurement counts. The next test applied noise during state preparation, which accumulates after each consecutive gate. Finally, the VQE optimizer itself received noisy cost values, and a recorded backend calibration was compared with the uniform test model. Throughout the experiments, the same $L = 4$ circuit and edge-string observable were used so that the source of each change remains clear.

The noiseless construction of the VQE and string-measurement circuit belongs to Zh. Liu's report, *Measuring Topology on Circuits*. The symmetry-based classification of physical Kitaev-chain errors and the scaling to large L are treated in Ed. Harutyunyan's report, *Error Taxonomy and Scaling to Large L*.

2 Noise model and diagnostic

2.1 Gate-level quantum channels

A quantum state is described by its density matrix ρ . An ideal gate g , represented by the unitary matrix U_g , changes it as $\mathcal{U}_g(\rho) = U_g \rho U_g^\dagger$. In the noisy model, a channel $\mathcal{E}_g$ acts immediately after the ideal gate:

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \quad \mathcal{E}_g(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I. \quad (2.1)$$

The matrices K_k are the channel's *Kraus operators*; each describes one possible effect of the noise, and k labels indices. The last equality is important, as it preserves the total probability, with I denoting the identity matrix. For a circuit of M gates, Eq. (2.1) is applied in gate order, starting from the input state ρ_0 . Noise then builds up during preparation instead of being added only to the completed state. Qiskit Aer performs this calculation with a density-matrix simulator and a NoiseModel [3].

The error test exploits a two-qubit depolarizing channel after every CNOT (cx) gate,

$$\mathcal{E}_p^{\text{dep}}(\rho) = (1 - p)\rho + p \frac{\text{Tr} \rho}{2^n} I. \quad (2.2)$$

Here p is the error strength and n is the number of qubits affected by the channel. In this report $n = 2$ because the channel follows a two-qubit CNOT. The state is unchanged with probability $1 - p$. The other term moves it towards a fully mixed n -qubit state. A second model uses thermal relaxation. For a gate lasting τ , energy relaxation has probability $p_{T_1} = 1 - e^{-\tau/T_1}$, where T_1 is the energy-relaxation time. The dephasing time T_2 describes the loss of phase coherence, which is known as decoherence. Relaxation can move the state out of the chosen parity sector, while dephasing weakens the coherence needed by the edge string. As mentioned before, the classification of these errors as physical Kitaev-chain errors is discussed in Ed. Harutyunyan's report on error taxonomy and large-chain scaling.

2.2 From a frozen state to noisy preparation

The initial isolation test was not a complete model of noisy state preparation. It started from a validated state that was kept fixed and applied symmetric readout error, local two-qubit depolarizing noise on nearest-neighbour pairs and finite-shot sampling. It did not include repeated noise during the preparation circuit, T_1/T_2 relaxation, angle miscalibration, or optimization with noisy cost values. This limited setup was useful because it showed how the diagnostic alone responds to controlled errors.

If ϵ is the probability of reading one qubit incorrectly, the readout error reduces an L -qubit edge string according to

$$\langle O_{\text{edge}} \rangle_{\text{readout}} = (1 - 2\epsilon)^L \langle O_{\text{edge}} \rangle, \quad (2.3)$$

where L is the chain length. Repeating the measurement N_{shots} times adds a statistical uncertainty that decreases roughly as $N_{\text{shots}}^{-1/2}$.

The next stage of the experiment produced the state with the noisy circuit itself. A channel acts after every relevant gate, so its effect depends on gate order and circuit depth. This is a better model of a device because errors alter the state while it is being built and accumulate with the number of gates. The final test also placed noise inside the VQE loop, so every trial set of angles was judged using a noisy circuit rather than an ideal one. This allows to remove the frozen-state assumption.

Angle errors form a separate case. If θ^* denotes the optimized gate angles, a calibration error uses $\theta^* + \delta\theta$. This coherent error rotates the state but keeps it pure. Random, or incoherent, channels generally lower the purity $\text{Tr} \rho^2$. Purity can therefore distinguish a wrong rotation from actual mixing even when both reduce $|\langle O_{\text{edge}} \rangle|$.

2.3 Expressibility versus accumulated noise

The preparation circuit uses a repeated layer, called an ansatz repetition. For r repetitions on L qubits, the linear circuit contains $r(L - 1)$ CNOTs. If p_{cx} is the error strength of each CNOT, a useful approximation to the measured signal is

$$|\langle O_{\text{edge}} \rangle|_{\text{meas}}(r) \simeq |\langle O_{\text{edge}} \rangle|_{\text{ideal}}(r) (1 - p_{\text{cx}})^{r(L-1)}. \quad (2.4)$$

The first factor states whether the circuit is deep enough to represent the target state. The second decreases the signal with every added layer. The optimal depth is therefore the shallowest circuit that can represent the state.

Putting noise inside the VQE changes the quantity being minimized. Let H be the qubit Hamiltonian, θ the trial gate angles, and $P = \prod_j Z_j$ the parity operator. The noisy cost is

$$C_{\text{noisy}}(\theta) = \text{Tr}[H\rho_\theta] + \lambda(1 - \text{Tr}[P\rho_\theta])^2, \quad \rho_\theta = \mathcal{E}\left(U(\theta)|0\rangle\langle 0|U^\dagger(\theta)\right). \quad (2.5)$$

Here $U(\theta)$ is the preparation circuit, $\mathcal{E}$ represents all noise applied through that circuit, and λ sets the penalty for leaving the target parity sector. A lower cost need not necessarily imply that the ground state was prepared more accurately. The new angles must therefore be run without noise and compared with the exact ground state.

3 Methods and Results

3.1 Circuit-level implementation and verification

All circuit results use a chain of $L = 4$ sites, hopping strength $t = 1$, pairing strength $\Delta = 1$, and the parity-constrained VQE and edge string from Zh. Liu’s report on noiseless circuit measurement. The chemical potential μ is varied to cross the phase transition. The preparation circuit contains single-qubit R_Y rotations and linear CNOT gates [2]. An error assigned to cx therefore acts after every CNOT. The output density matrix ρ gives the edge-string value $\text{Tr}(O_{\text{edge}}\rho)$.

The implementation was checked with a separate calculation that applies each gate and each depolarizing channel directly, without using Aer’s noise machinery. Across 41 values of μ/t , the two calculations differ by only about 10^{-15} . Figure 1 also compares the deliberately strong CNOT depolarizing strength $p_{\text{cx}} = 0.10$ with thermal relaxation at $T_1 = 80 \mu\text{s}$, $T_2 = 60 \mu\text{s}$ and CNOT duration of $\tau_{\text{cx}} = 300 \text{ ns}$. The relaxation probability per gate is only $1 - e^{-\tau_{\text{cx}}/T_1} \simeq 0.0037$, so the thermal curve remains closer to the ideal result, as can be seen.

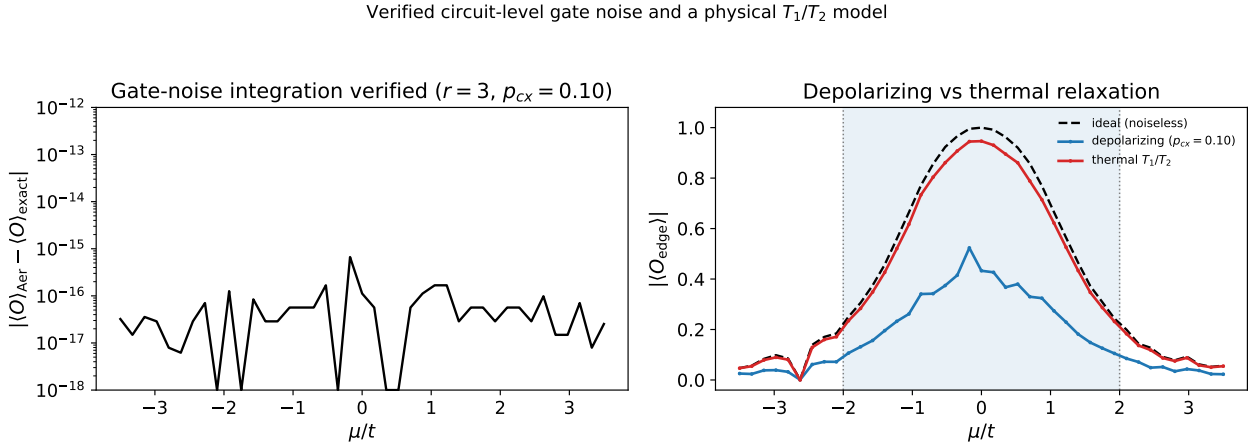


Figure 1: Left: the Aer NoiseModel edge string and the independent exact gate-by-gate reference agree to $\sim 10^{-15}$ across the μ sweep. Right: depolarizing versus thermal T_1/T_2 noise at matched gate placement.

3.2 The optimal depth is an expressibility threshold

A deeper VQE circuit can describe more states, but it also contains more noisy gates. The purpose of this test is to find the first depth that is rich enough to prepare the target state and to check whether any later layers still help after gate noise is included. This matters experimentally because naively choosing a depth from noiseless accuracy alone can produce a worse measured signal.

The test uses $\mu = 0$, where the ideal edge string is the strongest, so that any loss of signal is clearly caused by the circuit rather than by being close to the phase transition. Repetitions $r = 1, \dots, 6$ cover a circuit below the required depth, followed by the point where preparation first succeeds, and four deeper cases. The uniform CNOT error $p_{\text{cx}} = 0.05$ is an intentionally strong stress test: it is larger than the roughly 0.009–0.014 rates in the recorded backend used later, but small enough so that the signal remains visible across all six depths. It is chosen this way to expose the depth trend.

The noiseless edge string is only $\simeq 0.004$ at $r = 1$, reaches a value of approximately one at $r = 2$, and changes little in deeper circuits. The first circuit is therefore too shallow to represent the state. With

noise included, the signal peaks at $r^* = 2$ and so every further repetition adds three noisy CNOTs without improving the ideal state. This establishes a practical rule: we should choose the shallowest depth that passes the noiseless preparation check and then verify it under noise.

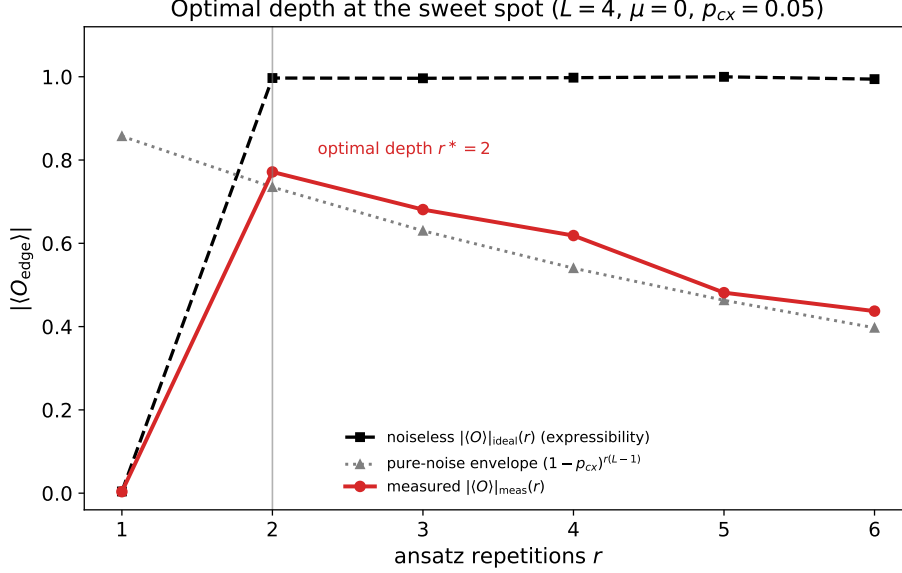


Figure 2: For $L = 4$, the noiseless edge string has an expressibility threshold: $r = 1$ fails, $r = 2$ succeeds, and deeper circuits mainly accumulate CNOT noise.

3.3 Noisy optimization and backend calibration

The preceding test used angles optimized without noise. Here the optimizer receives noisy energy values, testing whether it can adapt without merely making the edge string look better. We therefore compare both the noisy edge string and the noiseless fidelity to the exact ground state. The calculation fixes $L = 4$, $\mu = 0$, $\lambda = 0.1$, and $r = 3$. The depth is intentionally held one layer above the $r = 2$ preparation threshold in order to compare the error channels on the same capable circuit. This does not mean that we have a new depth recommendation.

For the testing, we let both depolarizing and amplitude-damping channel strengths run from 0 to 0.12 in steps of 0.02. This range allows to make a controlled comparison between two very different types of errors. Depolarizing noise, with two-qubit strength p_{CX} , is *unital*: it favours no state. On the other hand, amplitude damping with single-qubit probability γ is *non-unital*: it drives the state towards $|0\rangle$ and can shift the best angles.

channel	strength	$ \langle O_{\text{edge}} \rangle $ noisy		fidelity to exact GS	
		fixed θ^*	noise-aware	fixed θ^*	noise-aware
depolarizing	0.08	0.502	0.501	0.998	1.000
amplitude damping	0.08	0.422	0.565	0.998	0.994
amplitude damping	0.12	0.266	0.337	0.998	0.966

Table 1: Best-case versus noise-aware optimization at $L = 4$, $r = 3$, $\mu = 0$. Under amplitude damping the measured edge string rises while the noiseless fidelity to the exact ground state falls at strong damping.

Under depolarizing noise of strength 0.08, optimization leaves the edge string unchanged (0.502 com-

pared to 0.501) and the fidelity near one, supporting the idea of using ideal angles for this channel. Under amplitude damping, however, it raises the edge string from 0.422 to 0.565 at the noise strength of 0.08, yet the fidelity falls from 0.998 to 0.966 at 0.12. The optimizer can therefore improve the displayed signal without preparing a better state. This leads us to the conclusion that noisy VQE results need an independent state-quality check.

To check whether the uniform $p_{\text{cx}} = 0.05$ model is realistic, we repeat the same circuit using recorded calibration data from FakeManilaV2. This model assigns different error rates to individual gates, with CNOT rates ranging between 0.009 and 0.014. At $\mu = 0$, the ideal edge string is found to be 0.99, but falls to 0.87 with the Manila calibration and to 0.67 with uniform noise. The device snapshot therefore predicts noticeable, but less severe, signal loss than the stress test. In addition, it is worth noting that it is not a live hardware run, and readout error is excluded, which makes the value of 0.87 slightly optimistic.

To summarize, the central result is that noise-aware optimization and better state preparation are not equivalent. The large- L consequences are treated separately in Ed. Harutyunyan's report.

4 On the AI collaboration

My experience using large language models throughout the project was generally positive. We used Anthropic's Claude Opus 4.8 through Claude Code and OpenAI's GPT-5.6 Sol through Codex CLI. The models helped us to prepare the weekly presentations, write topic-specific notes, and solve the main coding challenges. They proved to be helpful as editors and as personal physics teachers, especially when generating all the learning materials. In addition, a collaborative GitHub repository and regular group meetings also contributed to a professional working environment. The project inspired me to learn more about quantum computing and recent developments in topological quantum computers.

References

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